

Don Hong

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SUMMARY

- Technical Knowledge: Embedded/Firmware, SSD/MFG Firmware, HW system Interface (PCIe/NAND/etc), HW/FW Debugging
- Proficient Tools: Embedded C/C++, Cortex-M, Xtensa, Trace32 (Lauterbach), NVMe, PCIe, Python, Linux, Git/Perforce/Jira

EXPERIENCE

Apple Inc. , Cupertino, CA

Embedded Systems Engineer

Dec 2025 – Present

- Currently working in the RF-Hardware (Wireless) Chipset Integration group as an embedded system engineer
- Lead integration of new silicon into shipping consumer platforms - owning root-cause analysis of fundamental and interface level failures (PCIe, I2C, SPMI, RFFE) that surface when a new chip meets the full HW/SW system
- Drive the full cycle: Silicon/system bring up -> NPI -> MP - designing and leading factory test-node architecture: Build scenario and stress-based case studies to proactively latent silicon/integration defects before MP
- Debug at signal, bus, and system level with DSOs and protocol Analyzer: analyze and sign off on hardware safety margins (reliability vs performance/power)

SK Hynix Memory Solutions Inc. (Solidigm Inc.), San Jose, CA

Staff Engineer, Firmware Development

Jul 2022 – Nov 2025

- Worked in the next-generation(PCIe 6 target) production firmware team
- Performed new FPGA Image/FW bring-up ARM Cortex R8/R82/M0+ and PCIe/NVMe subsystem IP
- Developed using embedded C and C++ programming on bare-metal (pre/post-silicon)
- Performed Soc verification (e.g., multi-core interconnection I2C, SPI, DMA, PCIe Controller, and SRAM/DRAM bring-up in the early stage chip development phase with FPGA before RTL freeze),
- Conducted firmware performance tuning with the silicon validation team through the prototyping platform
- Implemented firmware algorithms flow through the internal simulation tool regardless of device performance
- Implemented the feature of an error reporting/handling scheme in the production firmware
- Designed/Implemented a performance-critical customer (Google/Meta) requirement features in NVMe 2.0
- Worked closely with the design engineer to perform root cause analysis

SK Hynix Memory Solutions Inc., San Jose, CA

Senior Engineer, Firmware Development

May 2021 – Jul 2022

- Designed, implemented, and launched a time-sensitive debugging feature based on NVMe telemetry protocol to qualify the customer's data center requirement policy in an ASIC environment (Microsoft, DELL, HPE)
- Implemented, tested, and debugged Xtensa-based FW of the primary data path in the NAND interface layer and designed and implemented media defense algorithms on the FPGA, and committed to hardware bring-up
- Worked with the standard lab equipment, including JTAG, Oscilloscope, Logic Analyzer, and Power Analyzer
- Designed and built a chip reliability system for high-speed interface testing coverage (V6/7 NAND, PCIe 3/4/5)
- Collaborated with the SoC/Signal Integrity team to build debug firmware features (e.g., shmoo testing, module health monitoring sequence regarding the high-speed interface engineering between SoC, DRAM, and NAND)

Engineer, Application Software Development

Oct 2018 – May 2021

- Led and integrated test automation script for a web-based platform to enable scalable testing capability and simultaneous analysis of chip reliability testing/ Deployed test platform SDK: various APIs to regulate temperature, monitored and scheduled all test scenarios and client (eSSD) status, automated all running UART-based sequences of test cases, facilitating the use of the massive chamber systems for other engineers.
- Developed in-house SSD test infrastructure system with third-party software through cross-functional collaboration with external vendors, internal IT, Security, and Network server teams.

EDUCATION

University of California, Berkeley - B.A in Applied Mathematics

Jun 2015 – Aug 2018